

Incidence Rank and Bounded Defect for Linear Hypergraphs with Cograph Line Graphs

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16 August 2026

Abstract

Let $N(H)$ be the edge–vertex incidence matrix of a finite linear s -uniform hypergraph whose line graph is a cograph, and define

$$R_s(H) = (s + 1) \operatorname{rank}_{\mathbb{R}} N(H) - s|E(H)|.$$

The inequality $R_s(H) \geq 0$ holds, with equality characterized by affine planes of order s on every row-dependent component. Under the corresponding finite-net completion hypothesis, connected systems with bounded R_s admit an explicit classification.

If F is connected, $R_s(F) = d \leq s - 1$, the completion hypothesis holds through deficiency d , and $s > (d - 1)^2$, then either $N(F)$ has full row rank and $|E(F)| = d$, or F lies in a unique affine plane of order s . In the latter case, for some $0 \leq h \leq d$, the system contains every line in $s + 1 - h$ parallel classes and exactly $d - h$ additional lines from the remaining h classes. Conversely, each such system has defect d .

Bruck’s theorem gives the classification for every fixed d and all sufficiently large s , while Metsch’s completion theorem covers $d = o(s^{1/3})$. As a global consequence, if the total defect is at most D in the completion range, deleting at most D edges leaves a vertex-disjoint union of finite nets and affine planes. The associated column defect and row leverage are determined explicitly as well.

1 Introduction

For a linear s -uniform hypergraph H , the identity

$$N(H)N(H)^{\top} = sI + A(L(H))$$

relates the real incidence rank to the spectrum of the line graph. When $L(H)$ is a cograph, this relation yields

$$(s + 1) \operatorname{rank}_{\mathbb{R}} N(H) \geq s|E(H)|.$$

The row-dependent equality components are affine planes of order s , equivalently Steiner 2-designs $S(2, s, s^2)$. Write P_4^s for a linear s -uniform hypergraph on four edges whose line graph is an induced four-vertex path. The rank inequality then implies

$$|E(H)| \leq \frac{s + 1}{s} |V(H)|$$

for every linear P_4^s -free s -graph, the extremal problem studied by Adak and Verma [1].

The central parameter is the integer slack

$$R_s(H) = (s + 1) \operatorname{rank}_{\mathbb{R}} N(H) - s|E(H)|.$$

The cograph join decomposition gives an exact recursion for R_s . A join node of defect at most d contains at least $s + 1 - d$ minimal balanced co-components; these form parallel classes on a common s^2 -point set and hence a net of deficiency at most d . Classical completion theorems then constrain the remaining edges.

Under the corresponding net-completion hypothesis, a connected row-dependent system with $R_s = d$ lies in a unique affine plane and contains $s + 1 - h$ complete directions together with $d - h$ lines from the remaining h directions, for some $0 \leq h \leq d$. The only alternative is full row rank, in which case the system has exactly d edges. Bruck's theorem supplies the completion hypothesis whenever

$$s > \frac{1}{2}(d-1)^4 + (d-1)^3 + (d-1)^2 + \frac{3}{4}(d-1),$$

and Metsch's theorem whenever

$$3s > 8d^3 - 18d^2 + 8d + 4.$$

These bounds yield the component classification, a bounded edit result, and explicit formulas for column defect and row leverage.

2 Cographs and the incidence-rank inequality

For an s -uniform hypergraph H , let $N(H)$ be the edge-vertex incidence matrix, with rows indexed by $E(H)$, after isolated vertices are removed. Define

$$R_s(H) = (s + 1) \operatorname{rank}_{\mathbb{R}} N(H) - s|E(H)|$$

and

$$C(H) = |V^+(H)| - \operatorname{rank}_{\mathbb{R}} N(H),$$

where $V^+(H)$ is the set of nonisolated vertices. The quantity $R_s(H)$ is the *incidence-rank defect*. The identity

$$(s + 1)|V^+(H)| - s|E(H)| = R_s(H) + (s + 1)C(H). \quad (1)$$

Let $N(H)^\dagger$ denote the Moore-Penrose pseudoinverse. If $P_H = N(H)N(H)^\dagger$ is the orthogonal projector onto the column space of $N(H) \subseteq \mathbb{R}^{E(H)}$, the row leverage of $e \in E(H)$ is

$$\ell_H(e) = (P_H)_{ee}.$$

Throughout, $\mathbf{1}$ denotes an all-ones vector of the dimension determined by context.

The line graph $L(H)$ has vertex set $E(H)$, with two vertices adjacent when the corresponding hyperedges intersect. A graph is a *cograph* if it has no induced path on four vertices. Seinsche's decomposition theorem states that every cograph with at least two vertices is disconnected or has disconnected complement [5].

Lemma 2.1 (Cograph decomposition). *If a graph G with at least two vertices has no induced P_4 , then G is disconnected or $\overline{G}$ is disconnected.*

Proof. Assume that G is connected. Fix $x \in V(G)$, let $L_1 = N_G(x)$, and let $L_2 = V(G) \setminus (L_1 \cup \{x\})$. A shortest path in G is induced, so P_4 -freeness implies that every vertex is at distance at most two from x . If L_2 is empty, then x is isolated in $\overline{G}$, and we are done.

Let C be a connected component of $G[L_2]$. It has a neighbor in L_1 . Every $y \in L_1$ is either complete or anticomplete to C . Indeed, if its adjacency changes along an edge uv of a path in C , with $yu \in E(G)$ and $yv \notin E(G)$, then x, y, u, v induce a P_4 . Let

$$A_C = \{y \in L_1 : y \text{ is complete to } C\};$$

this set is nonempty.

The sets A_C , over the components of $G[L_2]$, are linearly ordered by inclusion. Otherwise choose distinct components C, C' , vertices

$$y \in A_C \setminus A_{C'}, \quad y' \in A_{C'} \setminus A_C,$$

and points $z \in C, z' \in C'$. If $yy' \in E(G)$, then z, y, y', z' induce a P_4 ; if $yy' \notin E(G)$, then z, y, x, y' do.

Choose a component C for which $A = A_C$ is minimal. Then A is complete to every component of $G[L_2]$. It is also complete to $L_1 \setminus A$: if $a \in A, y \in L_1 \setminus A$, and $ay \notin E(G)$, choose $z \in C$; the vertices z, a, x, y induce a P_4 . Finally, A is complete to x . Hence every edge between A and $V(G) \setminus A$ is present. Both parts are nonempty, so $\overline{G}$ is disconnected. $\square$

Lemma 2.2. *A linear s -uniform hypergraph is P_4^s -free if and only if its line graph is a cograph.*

Proof. A copy of P_4^s gives an induced P_4 in the line graph. Conversely, suppose that four hyperedges induce a graph path. Consecutive edges intersect and nonconsecutive edges are disjoint. By linearity the three consecutive intersections are single vertices. No two of these vertices coincide, since that would make a nonconsecutive pair intersect. The four edges therefore form P_4^s . $\square$

Identify an edge e with its incidence vector $x_e \in \{0, 1\}^{V(H)}$. Uniformity and linearity give

$$x_e^\top x_f = \begin{cases} s, & e = f, \\ 1, & ef \in E(L(H)), \\ 0, & ef \notin E(L(H)). \end{cases}$$

Equivalently,

$$N(H)N(H)^\top = A(L(H)) + sI.$$

Lemma 2.3 (No universal transversal). *Let $\mathcal{S}$ be an $S(2, s, s^2)$ contained in a linear hypergraph H . If $f \in E(H) \setminus E(\mathcal{S})$, then f cannot meet every block of $\mathcal{S}$.*

Proof. The design has $s(s+1)$ blocks, and each point lies in $s+1$ blocks. If f met every block, linearity would force each block to meet f once. Counting these incidences gives

$$s(s+1) = \sum_{v \in f \cap V(\mathcal{S})} (s+1) \leq s(s+1).$$

Thus every point of f belongs to $V(\mathcal{S})$. Two points of f lie in a unique block of the design, and that block would meet f twice. Linearity would force the block to equal f , contrary to the choice of f . $\square$

2.1 The join-node rank formula

Let F be a nonempty edge family whose line graph G is a connected nontrivial cograph. Let

$$F = F_1 \dot{\cup} \cdots \dot{\cup} F_p, \quad p \geq 2,$$

where the F_i are the connected components of $\overline{G}$; Lemma 2.1 gives $p \geq 2$. Thus every edge in F_i meets every edge in F_j when $i \neq j$. Put

$$\begin{aligned} m_i &= |F_i|, \\ \mathcal{R}_i &= \text{span}\{x_e : e \in F_i\}, \\ \rho_i &= \dim \mathcal{R}_i, \\ \mathcal{D}_i &= \text{span}\{x_e - x_f : e, f \in F_i\}. \end{aligned}$$

Lemma 2.4 (Orthogonal join decomposition). *The spaces $\mathcal{D}_1, \dots, \mathcal{D}_p$ are mutually orthogonal, and*

$$\dim \mathcal{D}_i = \rho_i - 1.$$

Let $\mathcal{D} = \mathcal{D}_1 \oplus \cdots \oplus \mathcal{D}_p$. For $e \in F_i$, let c_i be the orthogonal projection of x_e onto $\mathcal{D}^\perp$. This is independent of e , and

$$\mathbf{1}^\top c_i = s, \quad c_i^\top c_j = 1 \quad (i \neq j).$$

If

$$z = |\{i : \|c_i\|^2 = 1\}|,$$

then

$$\boxed{\text{rank } N(F) = \sum_{i=1}^p \rho_i - \max\{z - 1, 0\}.}$$

Moreover, all c_i of norm one are equal.

Proof. For $i \neq j$, every row from F_i has inner product one with every row from F_j . Hence

$$(x_e - x_f)^\top x_g = 0 \quad (e, f \in F_i, g \in F_j).$$

Thus $\mathcal{D}_i \perp \mathcal{R}_j$, and the $\mathcal{D}_i$ are mutually orthogonal.

Every vector in $\mathcal{D}_i$ has coordinate sum zero. Conversely, if

$$y = \sum_{e \in F_i} \alpha_e x_e \in \mathcal{R}_i, \quad \mathbf{1}^\top y = 0,$$

then $\sum_e \alpha_e = 0$. Fixing $f \in F_i$, we obtain

$$y = \sum_{e \neq f} \alpha_e (x_e - x_f) \in \mathcal{D}_i.$$

Thus $\mathcal{D}_i$ is the kernel in $\mathcal{R}_i$ of the nonzero coordinate-sum functional, proving $\dim \mathcal{D}_i = \rho_i - 1$.

Rows in F_i differ by elements of $\mathcal{D}_i$, so they have a common projection c_i . Since every row in $\mathcal{R}_i$ is orthogonal to $\mathcal{D}_j$ for $j \neq i$,

$$x_e - c_i \in \mathcal{D}_i.$$

The coordinate-sum assertion and the stated inner products now follow from the incidence inner-product formula.

The total row space is the orthogonal direct sum

$$\mathcal{D} \oplus \text{span}\{c_1, \dots, c_p\}.$$

Suppose

$$\sum_{i=1}^p a_i c_i = 0, \quad S = \sum_i a_i.$$

Taking coordinate sums gives $S = 0$. Taking inner product with c_j and using $c_i^\top c_j = 1$ for $i \neq j$ gives

$$0 = a_j(\|c_j\|^2 - 1) + S.$$

Hence $a_j = 0$ unless $\|c_j\|^2 = 1$. If two c_i, c_j have norm one, then $c_i^\top c_j = 1$, and therefore $c_i = c_j$. The relation space among the c_i has dimension $z - 1$ when $z \geq 1$, and dimension zero when $z = 0$. Combining this with the dimension formula for the $\mathcal{D}_i$ and this orthogonal decomposition proves the stated rank formula. $\square$

A co-component F_i with $\|c_i\|^2 = 1$ will be called *balanced*.

Lemma 2.5 (Balanced co-components). *Every balanced co-component F_i satisfies*

$$\rho_i = m_i, \quad m_i \geq s.$$

If $m_i = s$, then its s edges are pairwise disjoint and

$$c_i = \frac{1}{s} \sum_{e \in F_i} x_e.$$

Proof. Fix a balanced component and abbreviate $c = c_i$, $m = m_i$. Since $x_e - c \in \mathcal{D}_i$ and $c \perp \mathcal{D}_i$,

$$c^\top x_e = \|c\|^2 = 1.$$

Let Q be the disjointness graph on F_i . It is connected because F_i is a complement component. Put $y_e = x_e - c$. By the incidence inner-product formula and $c^\top x_e = 1$,

$$y_e^\top y_f = \begin{cases} s - 1, & e = f, \\ 0, & e \neq f \text{ and } e \cap f \neq \emptyset, \\ -1, & e \cap f = \emptyset. \end{cases}$$

Thus the Gram matrix of the y_e is

$$(s - 1)I - A(Q).$$

The vector c lies in the affine hull of the rows in F_i : $x_e - c$ is a linear combination of row differences. Thus

$$c = \sum_{e \in F_i} \lambda_e x_e, \quad \sum_e \lambda_e = 1.$$

Equivalently, $\sum_e \lambda_e y_e = 0$, so the displayed Gram matrix is singular. It is positive semidefinite, hence every eigenvalue of $A(Q)$ is at most $s - 1$, and $s - 1$ is an eigenvalue. Since Q is connected, Perron–Frobenius theory gives a one-dimensional kernel for this Gram matrix, spanned by a vector with all entries of one sign. In particular

$$s - 1 \leq m - 1,$$

so $m \geq s$.

If $\sum_e a_e x_e = 0$, then coordinate sums give $\sum_e a_e = 0$, and therefore $\sum_e a_e y_e = 0$. A nonzero kernel vector of this Gram matrix has all coordinates of one sign, whereas $\sum_e a_e = 0$. Hence every a_e vanishes and the rows are independent.

If $m = s$, the connected graph Q has order s and spectral radius $s - 1$, so $Q = K_s$. The edges are pairwise disjoint. The kernel of $sI - J$ is spanned by the all-ones vector, which gives the claimed formula for c . $\square$

Lemma 2.6 (Parallel-class bound). *At a fixed join node, at most $s + 1$ balanced co-components have size exactly s .*

Proof. Let $F_1, \dots, F_a$ be the balanced co-components of size s . By Lemma 2.4, their vectors c_i are equal to a common vector c . By Lemma 2.5, every F_i consists of s pairwise disjoint s -sets and

$$c = \frac{1}{s} \sum_{e \in F_i} x_e.$$

Consequently every F_i partitions the same set U of s^2 points, and $c = s^{-1} \mathbf{1}_U$. Edges in distinct classes meet once.

For each i , let

$$\mathcal{W}_i = \text{span} \left\{ x_e - \frac{1}{s} \mathbf{1}_U : e \in F_i \right\}.$$

This space has dimension $s - 1$ and lies in $\mathbf{1}_U^\perp$. If $e \in F_i$, $f \in F_j$, and $i \neq j$, then

$$\left(x_e - \frac{1}{s} \mathbf{1}_U \right)^\top \left(x_f - \frac{1}{s} \mathbf{1}_U \right) = 1 - 1 - 1 + 1 = 0.$$

The $\mathcal{W}_i$ are mutually orthogonal subspaces of the $(s^2 - 1)$ -dimensional space $\mathbf{1}_U^\perp$. Hence

$$a(s - 1) \leq s^2 - 1 = (s - 1)(s + 1),$$

and $a \leq s + 1$. $\square$

Theorem 2.7 (Cograph incidence rank). *Let H be a finite linear s -uniform hypergraph, $s \geq 2$, whose line graph is a cograph. Then*

$$\boxed{(s + 1) \text{rank}_{\mathbb{R}} N(H) \geq s|E(H)|.}$$

Equality holds if and only if every edge-containing component of H is a Steiner system $S(2, s, s^2)$.

Proof. Write

$$\rho(H) = \text{rank } N(H), \quad \sigma(H) = (s + 1)\rho(H) - s|E(H)|.$$

We induct on $m = |E(H)|$. The cases $m = 0, 1$ are immediate, with $\sigma(H) = 1$ when $m = 1$.

If $L(H)$ is disconnected, distinct line-graph components have disjoint point supports. Rank and σ are additive, and the induction hypothesis applies.

Now suppose that $L(H)$ is connected and $m \geq 2$. Use the root join decomposition and notation above, and let $\mathcal{B}$ be the set of balanced indices, with $z = |\mathcal{B}|$. If $z \leq 1$, Lemma 2.4 and induction give

$$\rho(H) = \sum_i \rho_i \geq \frac{s}{s + 1} \sum_i m_i = \frac{sm}{s + 1}.$$

Suppose $z \geq 2$. Balanced components have full row rank by Lemma 2.5. Therefore

$$\rho(H) = \sum_{i \notin \mathcal{B}} \rho_i + \sum_{i \in \mathcal{B}} m_i - z + 1 \geq \frac{s}{s+1} \sum_{i \notin \mathcal{B}} m_i + \sum_{i \in \mathcal{B}} m_i - z + 1.$$

It remains to prove

$$\sum_{i \in \mathcal{B}} m_i \geq (s+1)(z-1).$$

Let a be the number of balanced components of size s . Every other balanced component has size at least $s+1$, and Lemma 2.6 gives $a \leq s+1$. Thus

$$\sum_{i \in \mathcal{B}} m_i \geq as + (z-a)(s+1) = (s+1)z - a \geq (s+1)(z-1).$$

This proves the incidence-rank inequality.

We classify equality by the same induction. At a disconnected line graph, equality holds precisely when it holds in every component. Suppose the line graph is connected. If $z \leq 1$, the exact formula gives

$$\sigma(H) = \sum_i \sigma(F_i).$$

Equality would force equality in every F_i . By induction, one F_i contains an $S(2, s, s^2)$. An edge in another joined co-component would meet every block of that design, contrary to Lemma 2.3. Hence equality is impossible.

If $z \geq 2$, the exact formula is

$$\sigma(H) = \sum_{i \notin \mathcal{B}} \sigma(F_i) + \sum_{i \in \mathcal{B}} m_i - (s+1)(z-1).$$

The estimate above gives

$$\sum_{i \in \mathcal{B}} m_i - (s+1)(z-1) \geq s+1-a \geq 0.$$

If equality holds, then $a = s+1$. These $s+1$ minimal balanced components are parallel classes on a common set U of s^2 points. The number of unordered point-pairs covered by their blocks is

$$s(s+1) \binom{s}{2} = \binom{s^2}{2}.$$

Linearity prevents double counting, so every pair of points of U lies in exactly one block. The blocks form an $S(2, s, s^2)$. Lemma 2.3 excludes any additional joined edge.

Conversely, if N is the incidence matrix of $S(2, s, s^2)$, then

$$N^T N = sI + J,$$

so N has full column rank s^2 . Since the design has $s(s+1)$ blocks, equality holds in the incidence-rank inequality. $\square$

Corollary 2.8 (Linear four-edge paths). *For every $s \geq 2$ and every finite linear P_4^s -free s -uniform hypergraph H ,*

$$|E(H)| \leq \frac{s+1}{s} |V(H)|.$$

Equality holds if and only if H has no isolated vertices and its edge-containing components are vertex-disjoint copies of $S(2, s, s^2)$.

Proof. By Lemma 2.2, the line graph is a cograph. Theorem 2.7 gives

$$\frac{s|E(H)|}{s+1} \leq \text{rank } N(H) \leq |V(H)|.$$

Equality forces equality in both inequalities, and the statement follows from Theorem 2.7. The converse is immediate. $\square$

3 Finite nets and defect one

Standard finite-geometric terminology is used [3], with all definitions and arguments needed here stated explicitly.

An a -net of order s is a linear s -uniform incidence structure on s^2 points with a parallel classes. Each class consists of s disjoint lines partitioning the point set, and lines from different classes meet exactly once. The case $a = s + 1$ is an affine plane. Put

$$d = s + 1 - a,$$

the deficiency of the net.

Theorem 3.1 (Finite-net spectrum and defect). *Let $\mathcal{N}$ be an a -net of order s , with incidence matrix N . Then:*

1. $L(\mathcal{N}) = K_{s,\dots,s}$, with a parts;
2. the spectrum of $NN^\top$ is

$$sa \ (1), \quad s \ (a(s-1)), \quad 0 \ (a-1);$$

3. every point has degree a , and the eigenvalues of $a^{-1}NN^\top$ are

$$s \ (1), \quad \frac{s}{a} \ (a(s-1)), \quad 0 \ (a-1);$$

- 4.

$$\text{rank } N = 1 + a(s-1);$$

5. every line B has row leverage

$$\ell_{\mathcal{N}}(B) = \frac{1 + a(s-1)}{as};$$

6. the defects are

$$\boxed{R_s(\mathcal{N}) = d,} \quad \boxed{C(\mathcal{N}) = (s-1)d,}$$

and

$$\boxed{(s+1)|V(\mathcal{N})| - s|E(\mathcal{N})| = s^2d.}$$

Proof. The adjacency spectrum of the complete a -partite graph with s vertices in every part is

$$s(a-1) \ (1), \quad -s \ (a-1), \quad 0 \ (a(s-1)).$$

Since $NN^\top = sI + A(L(\mathcal{N}))$, the first spectrum and the rank formula follow. Every point lies on one line from every parallel class, so the normalized Gram matrix is $a^{-1}NN^\top$. The Gram pattern

is invariant under every permutation within and among the equal parts, so the leverage diagonal is constant; its value is rank divided by the number as of rows. Finally,

$$R_s(\mathcal{N}) = (s+1)(1+a(s-1)) - s(as) = s+1-a = d.$$

The remaining identities follow from the definitions and (1). $\square$

Deleting one parallel class from an affine plane therefore costs one incidence-rank-defect unit, $s-1$ column-rank-defect units, and s^2 units of path Turán slack. This is the discrete defect scale behind the equality theorem.

A single edge also has incidence-rank defect one, so the hypothesis $|E(F)| \geq 2$ is necessary in the characterization below.

Theorem 3.2 (First positive cograph defect). *Let F be a connected linear s -uniform hypergraph whose line graph is a cograph, and suppose $|E(F)| \geq 2$. Then*

$$R_s(F) = 1 \iff F \text{ is an } s\text{-net of order } s.$$

Proof. Use the root join decomposition above. Let $\mathcal{B}$ be the balanced indices, $z = |\mathcal{B}|$, and let a count the balanced co-components of size s .

If $z \leq 1$, Lemma 2.4 gives

$$R_s(F) = \sum_i R_s(F_i).$$

If this sum were one, then, since the join has at least two nonempty co-components, some other co-component would have defect zero. By Theorem 2.7, it contains an affine plane. Any edge in the positive-defect co-component would meet every line of that plane, contrary to Lemma 2.3. Hence $z \geq 2$.

The exact rank formula gives

$$R_s(F) = \sum_{i \notin \mathcal{B}} R_s(F_i) + \sum_{i \in \mathcal{B}} m_i - (s+1)(z-1).$$

Both terms are nonnegative integers, and

$$\sum_{i \in \mathcal{B}} m_i - (s+1)(z-1) \geq s+1-a.$$

A zero value in the preceding lower bound would force $a = s+1$. Those minimal components already form an affine plane, and Lemma 2.3 excludes every further joined edge. The resulting defect would be zero, not one. Therefore the join contribution is one, and every nonbalanced term has defect zero. A nonbalanced component cannot actually occur, since it would contain an affine plane joined to an edge in a balanced component. Thus all root co-components are balanced.

The same lower bound gives $a \geq s$. The case $a = s+1$ again gives an affine plane and defect zero, so $a = s$. These s minimal components are s parallel classes on a common s^2 -point set U ; they form an s -net.

It remains to exclude another balanced co-component. Every edge B in such a component meets all s lines of each existing class. A single class partitions U , so the s -set B is contained in U and is a transversal of every class. Define a graph Q on U by joining two points not lying together on a net line. Each point has

$$s^2 - 1 - s(s-1) = s-1$$

neighbors in Q . A transversal containing a point x consists of x and all $s - 1$ of its neighbors in Q . Hence two distinct transversals cannot intersect. Every extra co-component is therefore a family of pairwise disjoint s -sets contained in the s^2 -point set U . It has at most s members, while Lemma 2.5 gives at least s . It therefore has size s , making it another minimal component and contradicting $a = s$. Thus no extra component exists.

Conversely, Theorem 3.1 with $a = s$ gives $R_s = 1$. $\square$

Remark 3.3. Theorem 3.2 is an integral stability statement, not an edit-distance theorem. A global stability result would also have to control how these defect units distribute through the cograph cotree.

4 Join-node defect formula

The join formula used in Theorem 2.7 gives an exact recursion for R_s . At a join node, retain the notation

$$E(F) = F_1 \dot{\cup} \cdots \dot{\cup} F_p, \quad m_i = |F_i|, \quad \rho_i = \text{rank } N(F_i),$$

and let $\mathcal{B}$ be the set of balanced indices from Lemma 2.4. Put

$$z = |\mathcal{B}|, \quad a = |\{i \in \mathcal{B} : m_i = s\}|.$$

Lemma 4.1 (Join-node defect formula). *At a join node,*

$$R_s(F) = \begin{cases} \sum_{i=1}^p R_s(F_i), & z \leq 1, \\ \sum_{i \notin \mathcal{B}} R_s(F_i) + J_s(F), & z \geq 2, \end{cases}$$

where

$$J_s(F) = \sum_{i \in \mathcal{B}} m_i - (s+1)(z-1) = s+1-z + \sum_{i \in \mathcal{B}} (m_i - s).$$

Moreover

$$J_s(F) \geq s+1-a \geq 0.$$

At a disjoint-union node, R_s is additive.

Proof. Lemma 2.4 gives

$$\text{rank } N(F) = \sum_i \rho_i - \max\{z-1, 0\}.$$

Substitution in the definition of R_s gives the two displayed formulas. If $z \geq 2$, every balanced child has at least s edges by Lemma 2.5, and every balanced child not counted by a has at least $s+1$ edges. Hence

$$\sum_{i \in \mathcal{B}} (m_i - s) \geq z - a,$$

which gives the lower bound for $J_s(F)$. Additivity at a union node follows because distinct line-graph components have disjoint point supports. $\square$

In particular, if $R_s(F) \leq d$ and a join node has $z \geq 2$, then

$$a \geq s + 1 - d. \quad (2)$$

The a minimal balanced children are parallel classes on one s^2 -point set by Lemma 2.6. They therefore form an a -net of deficiency at most d .

The same calculation gives the rank of an incomplete collection of parallel classes. This will be used after a net has been completed.

Lemma 4.2 (Rank and leverage in affine-plane subsystems). *Let Π be an affine plane of order s . Let F consist of nonempty subfamilies $\mathcal{A}_1, \dots, \mathcal{A}_p$ of distinct parallel classes of Π , with $m_i = |\mathcal{A}_i| \leq s$. Put*

$$M = \sum_{i=1}^p m_i, \quad z = |\{i : m_i = s\}|.$$

Then

$$\text{null } N(F) = \max\{z - 1, 0\}, \quad \text{rank } N(F) = M - \max\{z - 1, 0\}.$$

If $z \geq 1$, every row in a full class has leverage

$$1 - \frac{z - 1}{zs},$$

and every row in an incomplete class has leverage 1. If $z = 0$, all rows have leverage 1.

Proof. The line graph is $K_{m_1, \dots, m_p}$, so

$$N(F)N(F)^T = sI + A(K_{m_1, \dots, m_p}).$$

Let $x = (x_B)$ lie in the kernel of this Gram matrix and put

$$S = \sum_{i=1}^p \sum_{B \in \mathcal{A}_i} x_B.$$

Subtracting two kernel equations in the same class shows that $x_B = c_i$ on $\mathcal{A}_i$. The remaining equations are

$$(s - m_i)c_i + S = 0 \quad (1 \leq i \leq p).$$

If one class is full, then $S = 0$. The constants on incomplete classes vanish, while those on the z full classes satisfy one linear relation, so the nullity is $z - 1$. If no class is full, substitution into the definition of S gives

$$S = -S \sum_i \frac{m_i}{s - m_i},$$

hence $S = 0$ and all $c_i = 0$.

For $z \geq 1$, the kernel consists of class-constant vectors supported on the full classes whose class constants sum to zero. Its orthogonal projector has diagonal $(z - 1)/(zs)$ on each row of a full class and zero on every incomplete-class row. Subtracting from the identity gives the leverage formula. $\square$

5 Completion and the bounded-defect classification

A net of order s and deficiency h has $s + 1 - h$ parallel classes. We use two classical completion results. Bruck proved that if $s > (h - 1)^2$, then an embedding of such a net in an affine plane, when it exists, is unique and every transversal of the net is a line of that completion [2]. His completion theorem also gives existence under

$$s > P(h - 1), \quad P(x) = \frac{1}{2}x^4 + x^3 + x^2 + \frac{3}{4}x.$$

Metsch proved the sharper sufficient condition

$$3s > 8h^3 - 18h^2 + 8h + 4$$

for completion [4].

Definition 5.1 (Completion hypothesis). For $0 \leq d \leq s - 1$, write $\text{Comp}(s, d)$ if

1. every net of order s and deficiency at most d embeds in an affine plane of order s ;
2. $s > (d - 1)^2$.

Thus $s > P(d - 1)$ implies $\text{Comp}(s, d)$. The result of Metsch implies the same conclusion whenever

$$3s > 8d^3 - 18d^2 + 8d + 4.$$

Theorem 5.2 (Completion-range classification). *Let F be a nonempty connected linear s -uniform hypergraph whose line graph is a cograph. Suppose*

$$R_s(F) = d \leq s - 1$$

and $\text{Comp}(s, d)$ holds. Exactly one of the following occurs.

1. Full row rank.

$$|E(F)| = d, \quad \text{rank } N(F) = d.$$

2. Affine-plane subsystem. *There are a unique affine plane Π of order s and an integer h , $0 \leq h \leq d$, such that $F \subseteq \Pi$, F contains every line in $s + 1 - h$ parallel classes of Π , and F contains exactly*

$$M = d - h$$

additional lines from the remaining h classes. If $d > 0$, then $1 \leq h \leq d$; if $d = 0$, then $h = 0$ and $F = \Pi$.

Conversely, every system in either class has incidence-rank defect d .

Proof. The identity

$$R_s(F) = |E(F)| - (s + 1) \text{null } N(F)$$

shows that a full-row-rank system has $|E(F)| = d$. Conversely, if $|E(F)| = d \leq s - 1$, its incidence rows are independent. Indeed, the Gram matrix of any d rows is $sI + A$, where A is a 0-1 matrix with zero diagonal and every row sum at most $d - 1 < s$. Strict diagonal dominance makes this Gram matrix positive definite.

Assume now that the rows are dependent. We argue by induction on d . The case $d = 0$ is Theorem 2.7. At the root join, use the notation of Lemma 4.1. The first claim is $z \geq 2$.

Suppose $z \leq 1$. Then

$$d = \sum_i R_s(F_i).$$

No line-graph component of any F_i can have defect zero: by Theorem 2.7 it would contain an affine plane, while any edge in another root co-component would meet every line of that plane, contrary to Lemma 2.3. Hence every nonempty line-graph component occurring among the F_i has positive defect strictly smaller than d .

If some F_i had a row-dependent line-graph component C , the induction hypothesis would place C in a unique affine plane and would give it at least

$$s + 1 - R_s(C) \geq s + 1 - d \geq 2$$

full directions on a point set U . Choose an edge g in another root co-component. Since g meets all s lines of a full class of C , its s distinct intersection points exhaust g , so $g \subseteq U$. The co-component F_i cannot have a second line-graph component, since that component is point-disjoint from U while g must meet all of its edges. Thus $F_i = C$. The complement of the line graph of a subset of an affine plane is the disjoint union of its nonempty directions; it is disconnected because C has at least two full directions. This contradicts the definition of F_i as a root co-component. Therefore every F_i has full row rank. Lemma 2.4 then makes F full row rank, a contradiction. Hence $z \geq 2$.

By (2),

$$a \geq s + 1 - d \geq 2.$$

The a minimal balanced co-components form an a -net $\mathcal{N}$ on a common point set U . Put

$$h = s + 1 - a \leq d.$$

Every edge outside these classes meets all s lines of one retained class. Its s distinct intersections exhaust the edge, so the edge lies in U and is a transversal of $\mathcal{N}$.

The completion hypothesis embeds $\mathcal{N}$ in an affine plane Π . Since $s > (h-1)^2$, Bruck's uniqueness and transversal theorem makes Π unique and places every transversal of $\mathcal{N}$ on a line of Π . Thus every edge of F is a line of this same affine plane.

For a subset of an affine plane, the connected components of the complement of the line graph are precisely its nonempty directions. A balanced direction has at least s lines by Lemma 2.5; it therefore consists of all s lines of that direction. Hence the $a = s + 1 - h$ minimal balanced components are exactly the full directions of F . Let M be the total number of retained lines in the remaining h directions. By Lemma 4.2,

$$R_s(F) = s + 1 - a + M = h + M.$$

Since the left side is d , we have $M = d - h$.

Conversely, an affine-plane subsystem in the statement has $a = s + 1 - h$ full directions and $M = d - h$ additional rows. Lemma 4.2 gives $R_s = h + M = d$. The independent case was handled at the beginning. $\square$

Corollary 5.3 (Fixed and growing defect). *Fix D . If $s > P(D-1)$, Theorem 5.2 classifies every connected cograph line system with $R_s \leq D$. More generally, the classification holds for a defect $d = d(s)$ whenever*

$$3s > 8d^3 - 18d^2 + 8d + 4;$$

in particular it holds for $d = o(s^{1/3})$.

For $d = 1$, the geometric alternative in Theorem 5.2 is an affine plane with one parallel class deleted. This sharpens Theorem 3.2 in the completion range.

Corollary 5.4 (Global bounded-defect decomposition). *Let H be a finite linear s -uniform hypergraph with cograph line graph. Suppose*

$$R_s(H) \leq D \leq s - 1$$

and $\text{Comp}(s, D)$ holds. After isolated vertices are removed, H is the vertex-disjoint union of affine planes of order s , full-row-rank components, and affine-plane subsystems as in Theorem 5.2. The positive component defects sum to at most D .

Deleting at most D edges turns H into a vertex-disjoint union of finite nets and affine planes. Alternatively, deleting the full-row-rank components and completing every geometric component requires at most sD added lines. For one geometric component of defect d and parameter h , the exact number of added lines is

$$hs - (d - h) = h(s + 1) - d.$$

Proof. Distinct line-graph components have disjoint point supports, and R_s is additive. Apply Theorem 5.2 componentwise. In a geometric component, delete its $d - h$ partial direction lines to leave a net; in a full-row-rank component, delete all d edges. The total number deleted is at most the total positive defect.

For the completion statement, add every missing line from the h directions of a geometric component. This requires $hs - (d - h) = h(s + 1) - d \leq sd$ lines. Summing over components gives the claim. $\square$

6 Consequences of the classification

Corollary 6.1 (Column defect and leverage). *In the geometric alternative of Theorem 5.2, with $d > 0$,*

$$|V(F)| = s^2,$$

$$\text{rank } N(F) = 1 + (s + 1 - h)(s - 1) + (d - h),$$

and

$$C(F) = hs - d.$$

The extremal slack is

$$(s + 1)|V(F)| - s|E(F)| = s((s + 1)h - d).$$

Every partial-direction row has leverage 1, while every row in a full parallel class has leverage

$$1 - \frac{s - h}{s(s + 1 - h)}.$$

Proof. There are $a = s + 1 - h$ full directions and $M = d - h$ partial lines. Lemma 4.2 gives

$$\text{rank } N(F) = as + M - (a - 1) = 1 + a(s - 1) + M.$$

The formulas follow by substitution. The leverage values are the two cases of Lemma 4.2. $\square$

For fixed d , the possible column defects of a dependent connected component are

$$s - d, 2s - d, \dots, ds - d,$$

subject to realization of the corresponding partial directions in the unique affine completion. Rows of leverage one are exactly the partial-direction rows and therefore identify the edges that must be deleted to recover the net core.

Corollary 6.2 (Affine-plane rank certificate). *Assume the hypotheses of Theorem 5.2. If a connected component with incidence-rank defect d is row dependent, then an affine plane of order s exists. Conversely, deleting d parallel classes from an affine plane of order s gives a connected row-dependent cograph line system of defect d , for $0 \leq d \leq s - 1$.*

Consequently, at an order s for which no affine plane exists, a connected row-dependent cograph incidence system cannot lie in the Metsch completion range. If $d = R_s(F)$, then necessarily

$$3s \leq 8d^3 - 18d^2 + 8d + 4,$$

so along non-plane orders

$$R_s(F) = \Omega(s^{1/3})$$

for connected row-dependent systems.

7 Concluding remarks

The incidence-rank defect is integer valued and additive across line-graph components. Theorem 5.2 shows that, throughout the finite-net completion range, every dependent connected component is obtained from a single affine plane by retaining all lines in most directions and a bounded number of lines in the remaining directions. At each join node, the cotree decomposition identifies the complete directions, while rows of leverage one identify the additional lines.

Beyond the completion range, the affine-plane description requires additional hypotheses. A natural problem is to determine whether a connected cograph line system with small R_s must contain large net substructures and whether deleting $O(R_s)$ rows leaves a disjoint union of nets that are inclusion-maximal within the original system. The real-rank argument used here relies on positive semidefinite Gram matrices and Perron–Frobenius theory; finite-characteristic incidence rank is a separate problem.

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